Urba Zafar

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HONORS & AWARDS

- **Chancellor's Bronze Medal |** Awarded for outstanding academic performance in undergraduate.
- **Dean's List of Honors Award |** Secured a perfect **4.00/4.00 SGPA** in the 8th semester.
- **Ranked 3rd among 90 students |** Maintained an overall **CGPA of 3.63/4.00** in BS Bioinformatics.
- **IELTS Academic |** Achieved 7.5 Bands in IELTS, indicating CEFR C1 Proficiency Level [Certificate]

EDUCATION

National University of Science and Technology (NUST) — Islamabad, PK
Masters in Bioinformatics — Oct 2026 – July 2028

Government College University Faisalabad (GCUF) | HEC RECOGNIZED — Faisalabad, PK
Bachelor of Science in Bioinformatics — Nov 2021 – June 2025

- **CGPA:** 3.63/4.0 | **SGPA:** 4.00/4.00 | **Rank:** 3/90

RESEARCH

Integrative Omics and Molecular Modeling Lab | *Research Assistant* — August 2025 – Present
Department of Bioinformatics & Biotechnology, GCUF
Principal Invigilator: Dr. Muhammad Tahir ul Qamar

- Developing a biological database integrating copy number variations (CNVs), epigenetic markers, and 3D structural data for neurodegenerative disease.
- Designing schema and pipelines to enable efficient integration, retrieval, and analysis of multi-omics data.
- Aiming to provide a comprehensive resource to support biomarker discovery and therapeutic research in neurological disease.

Biocatalysis and Protein Engineering Research Group Lab | *Research Assistant* — Feb 2025 – July 2025
Department of Bioinformatics & Biotechnology, GCUF [Methodology | Research Report]
Principal Invigilator: Dr. Muhammad Rizwan Javed

- Conducted integrative analysis of differentially expressed genes (DEGs) and copy number variations (CNVs) in Alzheimer's and Parkinson's disease.
- Constructed protein–protein interaction networks, performed GO enrichment and pathway analysis (DAVID, KEGG) to uncover disease-relevant pathways.
- Proposed potential biomarkers and therapeutic targets to advance understanding of neurodegenerative diseases.

Department of Computer Science | *Research Assistant* — November 2025 – Present
University of Idaho
Principal Invigilator: Dr. Hasan Jamil

- Developed an explainable Graph Neural Network (GNN) framework to identify Alzheimer's disease associated gene modules from protein-protein interaction (PPI) networks.
- Implemented GraphSAGE and GCN models with perturbation-based explainability to rank and interpret key disease genes.
- Validated top-ranked genes through KEGG pathway enrichment, Gene Ontology (GO) analysis, and GWAS overlap, confirming biological relevance.
- Achieved predictive performance of 89.6% accuracy using network topology features alone (GraphSAGE).

Fratta Lab | *Research Intern* — July 2026 – Oct 2026
UCL Queen Square Institute of Neurology
Supervisor: Dr. Pietro Fratta

- Investigating cryptic circRNA formation resulting from TDP-43 dysfunction in Inclusion Body Myositis (IBM), a muscle disease, using RNA-seq from IBM and control muscle biopsies.

- Detecting circRNAs via back-splice junction identification and identifying candidates significantly enriched in IBM versus control samples.
- Cross-referencing IBM-enriched circRNAs against known TDP-43-dependent cryptic splicing events from ALS/FTD datasets to assess mechanistic overlap.
- Classifying candidates as CNS-shared or muscle-specific to prioritize targets for experimental validation as stable biomarkers of TDP-43 loss of function.

SELECTED PROJECTS

Engineering a multi-epitope vaccine candidate against Leishmaniasis using comprehensive Immunoinformatics methods [Research Report] Sep 2023 – Feb 2024

Tools: NCBI, VaxiJen v2.0, AllerTOP v2.0, ExPASy ProtParam, PSIPRED, SCRATCH Protein Predictor, UCSF Chimera, GRAMM-X, PDB, PDBsum, C-ImmSim.

- Designed a multi-epitope vaccine targeting *Leishmania* proteins (Gp63, Histone H1) using immunoinformatics approaches.
- Predicted HTL, CTL, and B-cell epitopes; evaluated antigenicity, allergenicity, and toxicity.
- Modeled the 3D structure, performed TLR2 docking, and validated stability through molecular dynamics simulations.

Absolute Brainologists Mar 2024 – Jun 2024

- Developed a responsive quiz website using HTML, CSS, and JavaScript with real-time scoring features.
- Implemented dynamic navigation across quiz categories (e.g., Science, Nature) and automated results display.
- Designed a clean, mobile-friendly UI with customizable quiz options for enhanced user experience.

TECHNICAL SKILLS

Programming Languages: Python, R, C/C++, SQL, JavaScript, HTML, CSS.
Frameworks & Libraries: Pandas, NumPy, matplotlib, Biopython, FastAPI.
Softwares & Platforms: Linux, Jupyter Notebook, TBtools, Adobe Illustrator, Photoshop, Visual Studio.
Scientific Skills: Scientific Writing, Data Visualization, Research Methodology, Statistical Analysis, Literature Review.
Machine Learning: XGBoost, Random Forest, LASSO Regression, Deep Neural Networks, SHAP
Bioinformatics Tools:

- **Genomics & Gene Analysis:** Galaxy, TBtools, Genomics, MEGA, SAMtools.
- **Protein Modeling & Structure:** PyMOL UCSF, Chimera.
- **Sequence Alignment & Phylogenetics:** BLAST, Clustal Omega, MUSCLE, BioEdit.
- **Network & Pathway Analysis:** Cytoscape, STRING, KEGG Mapper, Reactome.
- **Databases:** NCBI, UniProt, PDB, Ensembl, ENA, GEO, ArrayExpress, ClinVar, Answer ALS.

ACADEMIC CERTIFICATIONS

- Fundamentals of Data Science in Precision Medicine and Cloud Computing	Stanford University	Dec 2024
- Programming Basics	Stanford University	Aug 2024
- Cloud Computing	Stanford University	Sep 2024
- Research Ethics	Stanford University	July 2024
- Oxford–Nanopore Third Generation Sequencing Workshop	Government College University Faisalabad	Oct 2024

VOLUNTEER & LEADERSHIP

Writer's Club | Media Secretary Nov 2023 – July 2025

- Managed social media presence and assisted in organizing literary and cultural events.
- Helped organize and promote literary and cultural activities on campus.
- Coordinated basic media coverage for events and wrote short posts or caption.

Amal Academy | Career-Prep Fellow Dec 2024 – Mar 2025

- Completed training in professional communication, leadership, and career development.

- Managed production, finances, and distribution, supporting five local schools.
- Led sustainability initiatives reaching 100+ students through eco-awareness drives.